

MINIMUM MEMOIZATION NEED NOT LIE IN THE GLUSHKOV MFN SET

ABSTRACT. Berglund, van der Merwe, and le Roux conjectured that a minimum memoization set for a Thompson or Glushkov automaton can be chosen inside its Minimum Feedback Node set, equivalently that some ordering makes IAR^+ return a minimum set. We disprove the Glushkov assertions. For every $k \geq 2$, the Glushkov automaton of

$$\left(\left(a \mid \underbrace{a^+ \mid \cdots \mid a^+}_{k \text{ occurrences}} \right) (b \mid c) \right)^*$$

has MFN size $k + 1$, but its unique minimum memoization set has size 2 and is disjoint from MFN. The full MFN set is its only IDA-eliminating subset, so IAR^+ returns all MFN states for every ordering. Both gaps are unbounded. The Thompson assertions are not addressed.

Berglund, van der Merwe, and le Roux introduced the Minimum Feedback Node scheme and conjectured that some minimum memoization set is contained in MFN, equivalently that some ordering of MFN makes IAR^+ return a minimum set [1, Conjectures 1 and 2]. We refute the Glushkov clauses of both conjectures.

For an automaton A , let $\mu(A)$ be the minimum number of memoized states needed to eliminate infinite degree of ambiguity (IDA), and set

$$\mu_{\text{MFN}}(A) = \min\{|M| : M \subseteq \text{MFN}(A) \text{ and } M \text{ eliminates IDA}\}.$$

Theorem 1. *For every $k \geq 2$, let*

$$R_k = \left(\left(a \mid \underbrace{a^+ \mid \cdots \mid a^+}_{k \text{ occurrences}} \right) (b \mid c) \right)^*, \quad G_k = \text{Gl}(R_k).$$

Then $\text{MFN}(G_k)$ is the unique minimum directed feedback vertex set of G_k , and

$$\mu(G_k) = 2, \quad \mu_{\text{MFN}}(G_k) = |\text{MFN}(G_k)| = k + 1.$$

The unique minimum memoization set is disjoint from $\text{MFN}(G_k)$, and IAR^+ returns all $k + 1$ MFN states for every ordering.

Proof. We use the IDA characterization of van der Merwe et al. [2], as stated in [1, Theorem 1]: for a memoized state set M , IDA occurs if and only if

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there are distinct states p, q and a word $v \in \Sigma^+$ with paths

$$p \xrightarrow{v} p, \quad p \xrightarrow{v} q, \quad q \xrightarrow{v} q,$$

such that the displayed cycle at q contains no state of M . Only that cycle is required to avoid M .

As in [1], positive closure is primitive. We apply the Glushkov construction to the displayed syntax without algebraic simplification, so repeated alternatives are distinct positions. Let x_0 be the position of the plain a , let $x_1, \dots, x_k$ be the positions of the k occurrences of a^+ , let y_b, y_c be the positions of b, c , and put

$$X_k = \{x_0, x_1, \dots, x_k\}.$$

With initial state q_0 , the nonempty transition values of G_k are

$$\begin{aligned} \delta(q_0, a) &= \delta(y_b, a) = \delta(y_c, a) = X_k, \\ \delta(x_i, b) &= \{y_b\} & (0 \leq i \leq k), \\ \delta(x_i, c) &= \{y_c\} & (0 \leq i \leq k), \\ \delta(x_i, a) &= \{x_i\} & (1 \leq i \leq k). \end{aligned}$$

The final states are q_0, y_b, y_c .

Every directed feedback vertex set contains $x_1, \dots, x_k$, because each has a self-loop. After deleting these states, the two cycles

$$x_0 \longrightarrow y_b \longrightarrow x_0, \quad x_0 \longrightarrow y_c \longrightarrow x_0$$

remain. A single further vertex meets both only if it is x_0 . Conversely, deleting X_k leaves only q_0, y_b, y_c , with no transitions among them. Thus X_k is the unique minimum directed feedback vertex set. The Glushkov MFN algorithm returns a minimum feedback vertex set [1, Theorem 4], so

$$\text{MFN}(G_k) = X_k.$$

Memoizing $M_0 = \{y_b, y_c\}$ eliminates IDA. Every nonempty cycle avoiding M_0 is an a -self-loop at some x_i with $i \geq 1$, traversed one or more times, and therefore has label a^r for some $r \geq 1$. If $p \xrightarrow{a^r} p$, then $p = x_j$ for some $j \geq 1$, and the only path on a^r from x_j ends at x_j . It cannot also end at a distinct state q . Hence $\mu(G_k) \leq 2$.

Conversely, let M eliminate IDA and suppose $|M| \leq 2$. If $y_b \notin M$, then $|X_k| = k + 1 \geq 3$ permits a choice of $q \in X_k \setminus M$ and $p \in X_k \setminus \{q\}$. There are paths

$$p \xrightarrow{ba} p, \quad p \xrightarrow{ba} q, \quad q \xrightarrow{ba} q.$$

The last cycle visits only q and y_b , neither in M , and is an IDA witness. Therefore $y_b \in M$. The same argument with ca gives $y_c \in M$. Thus M_0 is the unique IDA-eliminating set of size at most two, and $\mu(G_k) = 2$.

Finally, let $M \subsetneq X_k$. Choose $q \in X_k \setminus M$ and $p \in X_k \setminus \{q\}$. The same three paths on ba form an IDA witness, because the cycle at q visits only q

and $y_b \notin X_k$. Hence no proper subset of X_k eliminates IDA. Since deleting X_k is acyclic, X_k does eliminate IDA, and

$$\mu_{\text{MFN}}(G_k) = k + 1.$$

Removing any state from the initial MFN set therefore reintroduces IDA, so IAR^+ removes no state, regardless of the ordering. $\square$

The strict inequality $\mu_{\text{MFN}}(G_k) > \mu(G_k)$ refutes the Glushkov part of Conjecture 1 in [1], while the order-independent IAR^+ conclusion refutes Conjecture 2 for Glushkov automata. Moreover,

$$\mu_{\text{MFN}}(G_k) - \mu(G_k) = k - 1, \quad \frac{\mu_{\text{MFN}}(G_k)}{\mu(G_k)} = \frac{k + 1}{2},$$

so the additive and multiplicative gaps are unbounded. The case $k = 2$ gives the explicit five-position counterexample

$$\left((a \mid a^+ \mid a^+)(b \mid c) \right)^*.$$

REFERENCES

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